The relationship between scoring and salary vary by NBA player position

Contributions

Introduction

The relationship between NBA salaries and on-court performance is a subject of interest in sports analytics. Lyons et al. (2015) conducted a multiple regression analysis on the 2013 – 2014 salaries and professional performance of 243 players—excluding those on rookie-scale contracts—and found that metrics such as points, rebounds, and assists were associated with salary, a result that supports the simultaneous adjustment for multiple performance indicators.

More recent research has focused on prediction. Fu et al. (2025) analyzed 464 players from the 2022 – 2023 season using 26 input features; they reported a positive correlation between scoring and salary, and employed ensemble and optimization methods. Xu and Xu (2026) examined 467 players from the same season using 49 features, applying ensemble models alongside SHAP analysis; their case-specific interpretations revealed that age and playing time contribute differently to various predictions. While this provides a context for considering multidimensional player data, neither the observed correlations nor the local contributions to predictions can directly account for the differences in slopes across positions after adjustment.

Therefore, we investigate whether the association between average points per game and average seasonal salary—based on data from 2016 to 2019—varies by primary position, while controlling for average age, minutes played, assists, rebounds, steals, blocks, and turnovers. The response variable is the player's average salary (in millions of USD) across seasons for which both performance data and salary information are available; per-game performance metrics are weighted by the number of games played. Multiple linear regression allows for thesimultaneous control of these variables, enabling us to address the research question through estimates of the "points × position" interaction term, 95% confidence intervals (CIs), and a joint partial F-test—analyses that cannot be performed using simple correlations or comparisons of position-based means.

This study examines conditional associations using historical data, departing from the original purpose of such data—predicting All-Star rankings. For sports analysts and readers interested in sports economics, this research offers insight into the relationship between performance and compensation—as well as the associated uncertainty—across different positions. The findings are not intended to determine "fair" pay for individuals, nor should they be interpreted as establishing causal effects.

Data description

The data is sourced from the "NBA PLAYERS 2016– 2019" Kaggle dataset by Ratto (2019). The author compiled this dataset—originally intended for a university project on predicting All-Star voting rankings—by integrating information from other Kaggle datasets, Basketball Reference, and HoopsHype. As this is a secondary compilation, the specific data collection and matching procedures used in the original sources have not been fully verified.

The raw dataset contains 1,408 player-season records, covering 660 individual players. After excluding 62 records with missing salary data, 1,346 records remain. Aggregating these by player yields 631 unique individuals (29 players were excluded due to the lack of any valid season data). Salary and age are represented by their mean values, while per-game performance metrics are weighted by the number of games played. A player’ s primary position is defined as the position played in the highest number of games; in the event of a tie, the position is determined alphabetically. For any given player, the same set of valid seasons is used for all variables.

Salary values are positive, approximately continuous, and right-skewed; the assumption of normality applies to the conditional errors. While aggregating by player reduces redundancy, errors may still be correlated among players on the same team or those sharing the same season. Given that offensive responsibilities vary by position—potentially altering the relationship between scoring (PTS) and salary—an interaction term (PTS × position) is included. This allows for position-specific slopes for scoring, and the model tests whether a common slope would suffice; however, this does not in itself confirm the existence of actual differences between positions.

Table 1. Summary of all model variables. N = 631; no missing values. Salary: in millions of USD; Age: in years; MP: minutes per game. PTS, AST, TRB, STL, BLK, and TOV represent points, assists, rebounds, steals, blocks, and turnovers per game, respect ively. C, PF, PG, SF, and SG denote center, power forward, point guard, small forward, and shooting guard, respectively. For Salary, AST, TRB, and BLK, the mean exceeds the median, and the maximum values are relatively high;this suggests the need to examine the right tail of the distribution for potential high-value outliers though such values cannot be definitively classified as outliers based solely on this summary.

Variable	Mean	SD	Median	Min	Max
Salary	5.432	6.408	2.709	0.057	33.301
PTS	7.807	5.644	6.488	0.000	31.869
Age	25.803	4.352	25.000	19.000	41.000
MP	18.530	8.742	18.090	0.700	36.773
AST	1.732	1.645	1.200	0.000	10.459
TRB	3.354	2.309	2.848	0.000	15.118
STL	0.584	0.382	0.500	0.000	1.937
BLK	0.371	0.377	0.266	0.000	2.411
TOV	1.023	0.746	0.838	0.000	5.058

Position，n (%)：C 118 (18.7)；PF 134 (21.2)；PG 123 (19.5)；SF 100 (15.8)；SG 156 (24.7)。

Ethics discussion

Based on criteria of provenance, transparency, and accuracy, this compilation possesses a preliminary level of verifiability: the authors have documented their sources, and professional performance data as well as salary figures can be cross-referenced against public records. However, potential errors may arise from the matching of multiple data sources, the alignment of specific seasons, or variations in how contract terms are defined; public availability does not guarantee accuracy. Notably, 87 entries in the raw dataset show an identical salary of $77,250; these require further verification against contract terms rather than being summarily discarded as errors.

This study relies exclusively on publicly available professional information and does not introduce any new private data. The ethical acceptability of this approach remains contingent upon the permissions granted by the original sources and the manner in which the data is utilized --- public availability does not equate to explicit individual consent. We have cited our sources, respected usage permissions, and maintaine record of our data-cleaning process. Furthermore, we acknowledge potential selection bias regarding missing seasons, limitations in contract and experience data, and the possibility of team-dependent effects. Our conclusions are strictly confined to the historical sample; they do not constitute causal claims, nor do we employ residual values to assess the intrinsic worth of individual players.

Preliminary results

The initial model includes eight numerical predictors, the "position" variable, and the interaction term "PTS × position," totaling 17 parameters, with "C" (center) set as the reference category. N = 631, R^2 = 0.663, adjusted R^2 = 0.654, and the residual standard deviation is 3.769 million. Table 2 displays only thecoefficients discussed below; other terms remain in the full model.

The slope for PTS for the "C" position is 0.503, while for "SF" (small forward), it is 0.503 + 0.322 = 0.825 (million per point per game). Among the four interaction terms, only the one for "SF" has a classical confidence interval (CI) that excludes zero; however, the inclusion of zero for the others does not imply that their respective PTS slopes are zero. The joint partial F-test yields F(4, 614) = 3.396 (p = 0.00925); given the questionable validity of the underlying assumptions, this should be viewed as an exploratory result rather than definitive evidence that all positions differ from one another. The positive estimates for PTS, AST, and TRB are generally consistent with the findings of Lyons et al. (2015) and align with the positive correlation between scoring and salary noted by Fu et al. (2025) --- though correlation is not equivalent to an adjusted slope. The negative estimate for MP (minutes played), with a CI that includes zero, does not imply that increased playing time leads to a salary reduction, nor does it directly refute the local SHAP results presented by Xu and Xu (2026)

The variation in residual spread shown in Figure 1 casts doubt on the assumption of constant variance, while deviations in the upper tail of the Q-Q plot call the assumption of normality into question. Furthermore, the pattern observed in the "Response vs. Fitted" plot—along with 82 negative predicted values—suggests that the model for the conditional mean may be misspecified. Figure 2 provides an additional examination of the residual structure and location-based differences; the remaining residual plots for individual predictors are included in the analysis documentation. Due to the potential for shared team or season effects, the assumption of error independence remains unverified. No corrections for these violations are implemented at this stage.

Term	Estimate	95% CI
PTS	0.503	[0.300, 0.705]
MP	-0.041	[-0.141, 0.060]
AST	1.019	[0.512, 1.526]
TRB	0.722	[0.396, 1.049]
PTS × PF	0.048	[-0.136, 0.231]
PTS × PG	0.185	[-0.015, 0.385]
PTS × SF	0.322	[0.120, 0.524]
PTS × SG	0.073	[-0.125, 0.272]

Response vs Fitted 、Residuals vs Fitted、Normal Q-Q。

Response vs Fitted	Residuals vs Fitted	Normal Q-Q

Residuals vs PTS、Residuals by Position

Residuals vs PTS	Residuals by Position

Plan

Begin by verifying the definition of salary, checking for duplicate entries among low-salary records, and ensuring accurate player matching; do not automatically remove observations based solely on the magnitude of their residuals. If the residual spread or deviations in the upper tail of a Q-Q plot are observed, consider using a log transformation for salary, or apply a Box-Cox transformation as a guide—do not rely exclusively on the right-skewness of the salary distribution to justify such a transformation. If predictor-residual plots exhibit curvature, consider incorporating quadratic terms, while also taking the underlying context into account.

Since "points scored" and "position" are central to the research, they must be retained; furthermore, if an interaction term is included, the corresponding main effects must also be kept. Decisions regarding other predictors should be informed by existing literature, correlation analysis, and appropriate partial F-tests, rather than relying solely on individual p-values. When performing model comparisons, ensure that only nested models—those sharing the same response variable and the same set of observations—are evaluated; for instance, models using raw salary and log-transformed salary cannot be compared directly. Following any transformation or the addition/removal of variables, re-examine the mean structure, constant variance, normality assumptions, and potential issues with multicollinearity, and re-interpret the model coefficients. Note that "age" cannot serve as a substitute for "league experience."

Bibliography

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